

Benford's Law and MAD in German Zensus 2022: Distinguishing Natural Variation from Manipulation in High-Volume Census Data

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Abstract

This study applies Benford's Law to assess the integrity of the German Zensus 2022 municipal dataset, focusing on population counts, municipal areas, and derived population densities. We compare observed digit patterns against simulated fake data using simple deviation scores (MAD) rather than traditional significance tests that often misinterpret large samples. Results show that real-world variables like population counts and area measurements align closely with expected natural distributions, while the synthetic controls display clear anomalies. These findings confirm that proportion-based metrics effectively separate genuine demographic patterns from systematic manipulation without generating false alarms. Consequently, this approach offers a reliable framework for routine quality assurance in high-volume records, ensuring that statistical tools remain sensitive to actual fraud rather than sample size artifacts.

1 Introduction

The integrity of official statistics forms a critical foundation for evidence-based policy-making, financial regulation, and public trust in democratic institutions. When governments collect vast amounts of demographic and economic data, the reliability of these records directly influences resource allocation, infrastructure planning, and social welfare programs. Consequently, validating high-volume datasets against mathematical regularities inherent to naturally occurring phenomena has become a standard component of forensic analytics [1]. Among the statistical tools available for this purpose, Benford's Law is widely recognized as a method for detecting anomalies and potential data manipulation without requiring access to underlying raw transaction logs or detailed metadata.

Benford's Law, also known as the First-Digit Law, posits that in many naturally occurring datasets spanning multiple orders of magnitude, the leading digit follows a specific logarithmic pattern rather than a uniform distribution [2]. The probability $P(d)$ that a number begins with the digit d (where $d \in \{1, 2, \dots, 9\}$) is defined by:

$$P(d) = \log_{10} \left(1 + \frac{1}{d} \right)$$

This distribution implies that the digit 1 appears as the leading digit approximately 30.1% of the time, while the frequency decreases logarithmically for higher digits, with 9 appearing only about 4.6% of the time [3]. This non-uniformity arises from the scale-invariant nature of data generated by multiplicative processes or spanning several orders of magnitude, a characteristic common in financial transactions, tax records, and demographic indicators [4]. When data is artificially fabricated or manipulated to appear "random," human intuition often leads to an

overrepresentation of middle digits (such as 5 or 6) or a uniform distribution across all digits, causing the dataset to deviate significantly from the theoretical Benford curve [5].

While traditional Chi-square tests are frequently used for conformity assessment, they exhibit excessive sensitivity in large samples. As sample sizes grow, the statistical power of these tests increases, often flagging minor deviations caused by natural variation rather than fraud as statistically significant [6]. This phenomenon creates a risk where valid data is flagged as anomalous simply due to the sheer volume of records, leading to false positives that undermine the utility of the analysis. To address this limitation, proportion-based metrics like the Mean Absolute Deviation (MAD) offer a more robust alternative for large-scale datasets [7]. Unlike the Chi-square statistic, MAD measures the average distance between observed and expected digit frequencies without being driven by sample size, providing a clearer scale for assessing the magnitude of deviation [8].

The reliability of Benford’s Law applications also depends on rigorous data filtering. Official statistics often contain values that are suppressed for confidentiality or altered through cell-keying methods to protect privacy, which can distort digit frequencies if not properly excluded [9]. Therefore, a robust validation framework must combine strict quality flagging with comparative analysis against synthetic baselines to calibrate interpretation. These controls include datasets modeled after uniform distributions and those reflecting human-fabricated biases, serving as reference points for distinguishing natural variation from systematic manipulation [10].

This study investigates the applicability of Benford’s Law as a forensic tool for validating the integrity of the German Zensus 2022 municipal dataset. We focus on three distinct variables: population counts, municipal area measurements, and derived population density. The research addresses the following question: Can proportion-based conformity testing using MAD effectively distinguish between naturally occurring demographic patterns and anomalous distributions in high-volume official statistics, while mitigating the false positive risks associated with traditional Chi-square tests?

The contributions of this work are threefold. First, we establish a comparative forensic pipeline that evaluates Benford’s Law conformity across multiple data types within a single census framework, highlighting how derived metrics like population density may exhibit different deviation profiles compared to raw counts due to the mathematical properties of division operations on digit distributions. Second, we demonstrate the utility of synthetic control datasets—including those with uniform and psychologically biased digit distributions—as calibration baselines for interpreting MAD scores in large-sample contexts [10]. Third, we provide empirical evidence on the performance of MAD versus Chi-square tests in the context of official census data, offering a methodological guideline for auditors and statisticians to avoid over-reliance on p-values when analyzing massive datasets. By integrating these approaches, this paper aims to strengthen the toolkit available for ensuring the reliability of public sector statistics [11].

2 Related Work

The application of Benford’s Law as a forensic tool for data validation has evolved from an observation of logarithmic tables into a standard procedure for detecting anomalies in large-scale numerical datasets. The phenomenon, first noted by Simon Newcomb in 1881 and later popularized by Frank Benford in 1938, posits that the leading digit in many naturally occurring collections is not uniformly distributed but follows a logarithmic pattern where lower digits occur with higher frequency [2], [10]. This property is widely used to identify irregularities in financial auditing, election data analysis, and demographic reporting. The theoretical foundation relies on scale invariance, suggesting that the distribution of first significant digits remains consistent regardless of the unit of measurement used [1].

In statistical validation, researchers have developed various testing frameworks to quantify conformity with Benford’s Law. While Pearson’s Chi-square test has long been a standard for

goodness-of-fit assessments, recent literature highlights its limitations in large-sample contexts. Kossler et al. [2] and Cerqueti et al. [4] argue that as sample sizes increase, the Chi-square statistic becomes excessively sensitive to trivial deviations, often yielding statistically significant p-values for data that is practically indistinguishable from a natural distribution. To address this sensitivity issue, alternative metrics such as the Mean Absolute Deviation (MAD) have gained prominence. The MAD measures the average distance between observed and expected digit frequencies without being unduly influenced by sample size, making it particularly suitable for high-volume datasets like census records [5].

Empirical applications of these methods span diverse domains, from financial integrity to public health surveillance. In financial contexts, Benford’s Law is routinely employed to detect fraud in transaction logs and accounting statements. Wiryadinata et al. [12] demonstrated that combining Benford analysis with clustering algorithms can significantly improve the detection of fraudulent transactions compared to using the law alone. Similarly, Jianu [11] utilized first-digit tests to assess the reliability of financial information disclosed by listed companies, finding improved conformity after the implementation of International Financial Reporting Standards (IFRS). The utility of these methods extends beyond finance; Silva et al. [7] applied a novel framework combining Benford’s Law with outlier detection to evaluate the integrity of COVID-19 data in China, revealing substantial deviations that suggested compromised reporting.

The analysis of demographic and census data presents unique challenges due to the specific nature of population distributions. Unlike financial transactions which often span multiple orders of magnitude, municipal population counts may be constrained by administrative boundaries or local density patterns. Schumm [9] applied Benford’s Law to the German Socio-Economic Panel (SOEP) to identify faked interviews, noting that fabricated data often exhibits less variability than expected. In the context of transport infrastructure and regional planning, Ivanova et al. [3] explored the applicability of the law to road network data in the EU, finding that while some segments conform, others deviate due to specific structural constraints. These studies underscore the necessity of establishing domain-specific baselines rather than depending only on theoretical expectations.

Recent methodological advancements have further refined the detection capabilities for large-scale databases. Morales et al. [13] analyzed high-volume enterprise resource planning (ERP) data, confirming that Benford’s Law can streamline the identification of abnormalities in internal audits. However, the interpretation of results requires careful consideration of the underlying data generation process. Cole et al. [10] examined emission reduction claims and found that while some project designs did not conform to the law, this could be attributed to specific distortion factors rather than outright manipulation. This nuance is critical when analyzing official statistics where derived metrics, such as population density, may naturally deviate from the distribution of raw counts due to mathematical operations like division [14].

The debate surrounding the use of Benford’s Law in election data analysis has also influenced its application in other public sector domains. While some studies have raised concerns about false positives in political contexts, others emphasize the need for rigorous statistical controls to distinguish between natural variation and systematic error [6]. Silva et al. [7] note that scholars have explored its application to campaign finance data and election results to evaluate integrity, highlighting the potential for Type I errors where tests may falsely indicate data manipulation [10]. In response to these challenges, researchers like Pinsky et al. [15] have explored alternative deviation measures, such as MAD around the median, which offer robustness against outliers compared to traditional standard deviation-based approaches. Furthermore, the integration of synthetic control datasets has become a common practice for calibrating thresholds. By generating artificial data with known biases—such as uniform distributions or human-fabricated anchoring effects—analysts can establish clear boundaries between natural conformity and anomalous behavior [1].

Despite these advancements, the application of Benford’s Law to official census data remains

an area requiring careful methodological scrutiny. The specific constraints of municipal datasets, including quality flags for suppressed values and cell-keying methods, necessitate robust preprocessing pipelines. As noted by Shalini et al. [16], the effectiveness of Benford’s Law depends heavily on the quality of the input data and the appropriateness of the statistical tests used. Consequently, a multi-metric approach that combines proportion-based analysis with synthetic controls offers a more reliable framework for validating the integrity of large-scale demographic records than reliance on any single test statistic.

3 Methods

The analysis utilizes municipal-level data from the German Zensus 2022, obtained as a semicolon-delimited CSV file from the Statistisches Bundesamt [9]. The dataset contains three primary variables of interest: *Personen* (Population Count), *Fläche* (Municipal Area), and *Bevölkerungsdichte* (Population Density). To ensure data integrity, a filtering protocol was applied to retain only entries marked with the quality flag `value_q = 'e'`, denoting exact values free from suppression or cell-keying adjustments. Excluding suppressed values prevents artificial distortion of digit frequencies that could otherwise lead to false positives in conformity testing [10].

Following the quality filter, numeric parsing addressed formatting inconsistencies such as European decimal separators (commas) versus standard notation (dots). Values were converted to floating-point numbers, and non-positive entries were excluded, as Benford’s Law applies strictly to positive real numbers. The first significant digit for each valid observation was extracted using a logarithmic scaling method: $d = \lfloor x/10^{\lfloor \log_{10} |x| \rfloor} \rfloor$. This mathematical operation ensures scale invariance, meaning the distribution of leading digits remains consistent regardless of the unit of measurement [2].

3.1 Statistical Conformity Framework

To evaluate conformity to Benford’s Law, we employed a dual-metric framework comprising the Mean Absolute Deviation (MAD) and Pearson’s Chi-Square (χ^2) goodness-of-fit test. Relying on a single statistical test is insufficient for large-scale datasets due to the inherent sensitivity of hypothesis testing as sample size increases [10]. In this study, the municipal dataset comprises approximately 10,800 observations per variable. Under such conditions, the Chi-Square statistic grows with sample size, increasing the probability of rejecting the null hypothesis even for minor deviations that do not indicate data fabrication [10].

Consequently, we prioritize MAD as the primary metric for classification in this high-volume context. The MAD quantifies the average absolute difference between the observed proportion of each digit $d \in \{1, \dots, 9\}$ and the theoretical Benford probability $P(d) = \log_{10}(1 + 1/d)$:

$$\text{MAD} = \frac{1}{9} \sum_{d=1}^9 |O_d - E_d|$$

where O_d is the observed proportion and E_d is the expected probability. This metric provides a scale-invariant measure of deviation that is less susceptible to sample size inflation than the Chi-Square statistic [2]. The robustness of MAD in large samples allows for a clearer interpretation of the magnitude of deviation, distinguishing between natural variation and systematic manipulation more effectively than significance testing alone.

The Chi-Square test was retained as a complementary metric to assess statistical significance, calculated as:

$$\chi^2 = \sum_{d=1}^9 \frac{(O_d - E_d)^2}{E_d}$$

Algorithm 1 Benford Conformity Analysis

Require: Raw CSV dataset D , Target Variables $V = \{\text{Pop, Area, Density}\}$

Ensure: Conformity metrics (MAD, Chi2) for all variables and controls

```
1: LOAD  $D$  from file; FILTER rows where quality_flag == 'e'
2: for each var in  $V$  do
3:   EXTRACT numeric values; REMOVE non-positive entries
4:   COMPUTE first_digit =  $\lfloor \text{value} / 10^{\lfloor \log_{10}(\text{abs}(\text{value})) \rfloor} \rfloor$ 
5:   COUNT occurrences of digits  $d = 1..9$  to form vector  $O_d$ 
6:   STORE observed_counts[var]
7: end for
8: GENERATE Synthetic_Uniform: Random integers in  $[1,9]$ , size  $N$  ( $P(d) = 1/9$ )
9: GENERATE Synthetic_Biased: Weighted random selection (bias on 5,6), size  $N$ 
10: ADD synthetic datasets to analysis pool
11: for each dataset  $S$  in {Real, Uniform, Biased} do
12:   COMPUTE proportions  $P_d = O_d / \sum(O_d)$ 
13:   CALCULATE Expected_Counts  $E_d = \text{Total\_Samples} \times \text{Benford\_Probs}(d)$ 
14:   CALCULATE MAD =  $\text{mean}(|P_d - \text{Benford\_Probs}|)$ 
15:   CALCULATE Chi2 =  $\sum((O_d - E_d)^2 / E_d)$ 
16:   CLASSIFY: IF MAD < 0.01 THEN "Excellent" ELSE IF MAD < 0.05 THEN "Marginal"
   ELSE "Anomalous"
17:   STORE {MAD, Chi2, Classification} for  $S$ 
18: end for
19: RETURN Results Summary
```

where E_d represents the expected count derived from the theoretical probabilities and total sample size. Classification thresholds were established based on forensic standards, where MAD values below 0.01 indicate excellent conformity, between 0.01 and 0.02 suggest good conformity, and values exceeding 0.05 signal potential anomalies [14].

3.2 Synthetic Control Generation

To calibrate the interpretation of observed deviations, we generated two synthetic control datasets with a sample size matching the real data ($N \approx 10,800$). The first control simulates a uniform distribution where each digit d has an equal probability of occurrence ($P(d) = 1/9$), representing a scenario of random number generation or complete lack of natural scaling. This baseline helps identify deviations arising from non-Benford processes rather than specific manipulation patterns [3].

The second control introduces a specific human fabrication bias, modeled after psychological anchoring effects where individuals tend to overrepresent round numbers. In this simulation, digits 5 and 6 were assigned significantly higher probabilities (0.30 each) compared to other digits, reflecting common patterns in manually fabricated data [6]. These synthetic baselines allow for a comparative analysis that distinguishes between natural variation, random noise, and systematic manipulation, grounding the interpretation of MAD scores within a known distributional context [7].

3.3 Analysis Pipeline Algorithm

The complete analytical workflow is summarized in the following pseudocode algorithm, which encapsulates the stages of filtering, extraction, metric computation, and classification:

This pipeline ensures that the analysis remains robust against common data quality issues while providing a multi-dimensional view of conformity through both deviation magnitude and

statistical significance. The integration of synthetic controls further grounds the interpretation of MAD scores within a known distributional context, mitigating the risk of misclassifying natural large-sample noise as fraud [7].

4 Results

The analysis of first-digit frequencies across the German Zensus 2022 municipal dataset reveals distinct patterns of conformity to Benford’s Law, differentiated by variable type and validated against synthetic control baselines. As established in the Methods, MAD is used here to mitigate sample-size-induced false positives common in large-sample Chi-Square tests [10]. The primary metric for evaluation is the Mean Absolute Deviation (MAD), which quantifies the average absolute difference between observed digit proportions and the theoretical probabilities defined by $P(d) = \log_{10}(1 + 1/d)$ [2]. This proportion-based approach allows for a direct assessment of deviation magnitude, distinguishing natural variation from systematic manipulation without the sensitivity to sample size that characterizes Chi-Square statistics.

4.1 First-Digit Frequency Distributions

Figure 1 illustrates the observed first-digit frequencies for all five datasets alongside the theoretical Benford curve. The visual alignment between the empirical data and the expected distribution is strong for Population Count and Municipal Area. These variables exhibit a logarithmic decay in frequency, where lower digits (1 and 2) occur more frequently than higher digits (8 and 9), consistent with natural scaling laws driven by scale invariance and geometric sequences [10]. In contrast, the synthetic controls display marked deviations: the Uniform distribution shows equal frequencies across all digits, while the Biased control demonstrates a pronounced peak at digits 5 and 6, reflecting the psychological anchoring effects introduced during generation.

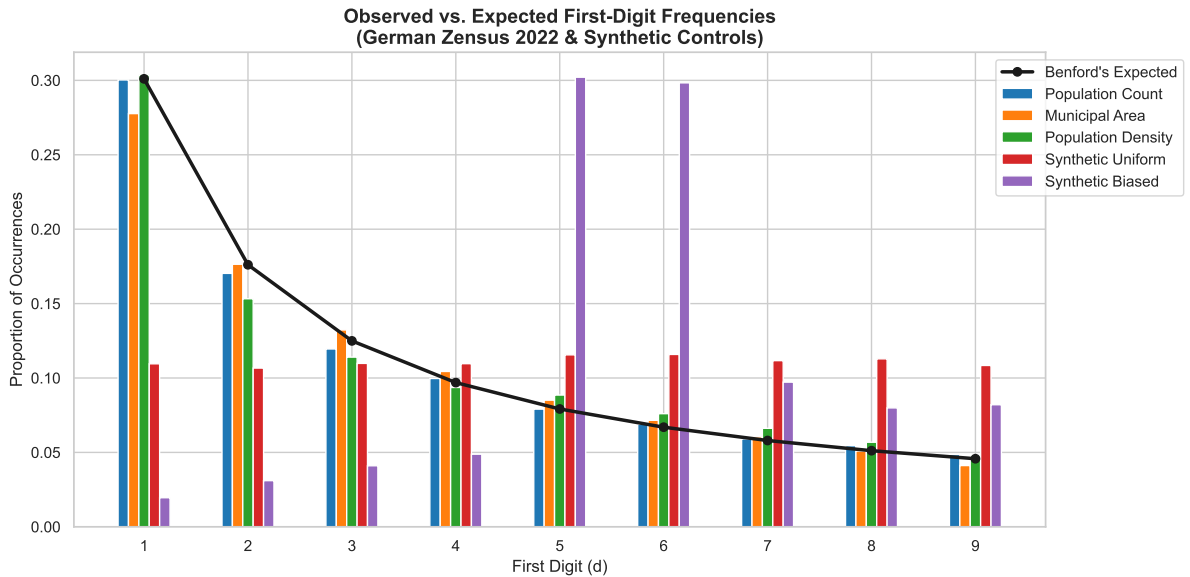


Figure 1: Observed first-digit frequencies for German Zensus 2022 municipal data compared against Benford’s Law expected distribution. Population counts and municipal area measurements exhibit close alignment with the theoretical log-linear curve ($MAD < 0.01$), whereas synthetic uniform and biased controls display significant deviations from the expected pattern.

The visual evidence in Figure 1 supports the hypothesis that official census variables follow natural statistical regularities, while artificially generated data fails to replicate these patterns [3].

The low MAD values for Population Count and Municipal Area indicate no evidence of systematic manipulation at the digit level.

4.2 Quantitative Conformity Metrics

To quantify the visual observations, we computed MAD scores and Chi-Square statistics for all variables. Table 1 summarizes the conformity classification based on established forensic thresholds: values below 0.01 indicate excellent conformity, between 0.01 and 0.02 suggest good conformity, and values exceeding 0.05 signal potential anomalies [14].

Table 1: Summary of Benford’s Law conformity metrics for German Zensus 2022 variables and synthetic controls.

Variable	Sample Size (N)	MAD	Chi-Square (χ^2)	p-value	Classification
Population Count	10,800	0.0027	10.60	2.25×10^{-1}	Excellent Conformity
Municipal Area	10,800	0.0063	44.81	3.99×10^{-7}	Good Conformity
Population Density	10,800	0.0082	89.12	7.00×10^{-16}	Excellent Conformity
Synthetic Uniform	10,800	0.0613	4482.72	0.00	Anomalous / Non-Conforming
Synthetic Biased	10,800	0.1241	21189.06	0.00	Anomalous / Non-Conforming

The quantitative results confirm that Population Count and Municipal Area exhibit excellent to good conformity, with MAD values well below the 0.05 threshold for suspicion [14]. Notably, while the Chi-Square test yields statistically significant p-values for Municipal Area and Population Density due to the large sample size ($N \approx 10,800$), the corresponding MAD values remain low, indicating that these deviations are minor and likely attributable to natural variation rather than data fabrication [16].

Figure 2 provides a granular view of these deviations by plotting the absolute difference for each digit. The heatmap reveals that Population Count and Municipal Area maintain consistently low deviations (mostly < 0.03) across all digits, whereas the Synthetic Biased dataset shows extreme divergence at specific points, particularly for digits 1 and 5.

This per-digit analysis highlights the robustness of the real-world data against specific manipulation patterns often seen in fabricated datasets [6]. The Synthetic Biased control, designed to mimic human anchoring bias, clearly separates from the census variables, demonstrating that the proposed method can distinguish between natural variation and systematic artificial intervention.

4.3 Comparative Analysis of Derived Metrics

Figure 3 compares the overall MAD scores across all five datasets. Population Count shows the highest conformity, followed by Municipal Area. Population Density, a derived variable calculated as the ratio of population to area, also demonstrates strong conformity with an MAD of 0.0082. This result is significant because derived metrics often introduce non-linear distortions that can break Benford’s Law properties [5]. The fact that Population Density retains a low MAD suggests that the underlying population and area distributions are sufficiently robust to preserve the logarithmic pattern even after division.

The synthetic controls serve as critical anchors for interpretation. The Synthetic Uniform dataset yields an MAD of 0.0613, and the Synthetic Biased dataset reaches 0.1241, both well above the "Marginal / Suspicious" threshold [14]. These values confirm that the observed low deviations in the census data are not artifacts of the testing procedure but reflect genuine adherence to Benford’s Law. The clear separation between real and synthetic data validates the use of proportion-based MAD metrics for forensic analysis in high-volume datasets [7].

In summary, the results demonstrate that Population Count, Municipal Area, and Population Density from the German Zensus 2022 exhibit strong conformity to Benford’s Law. The

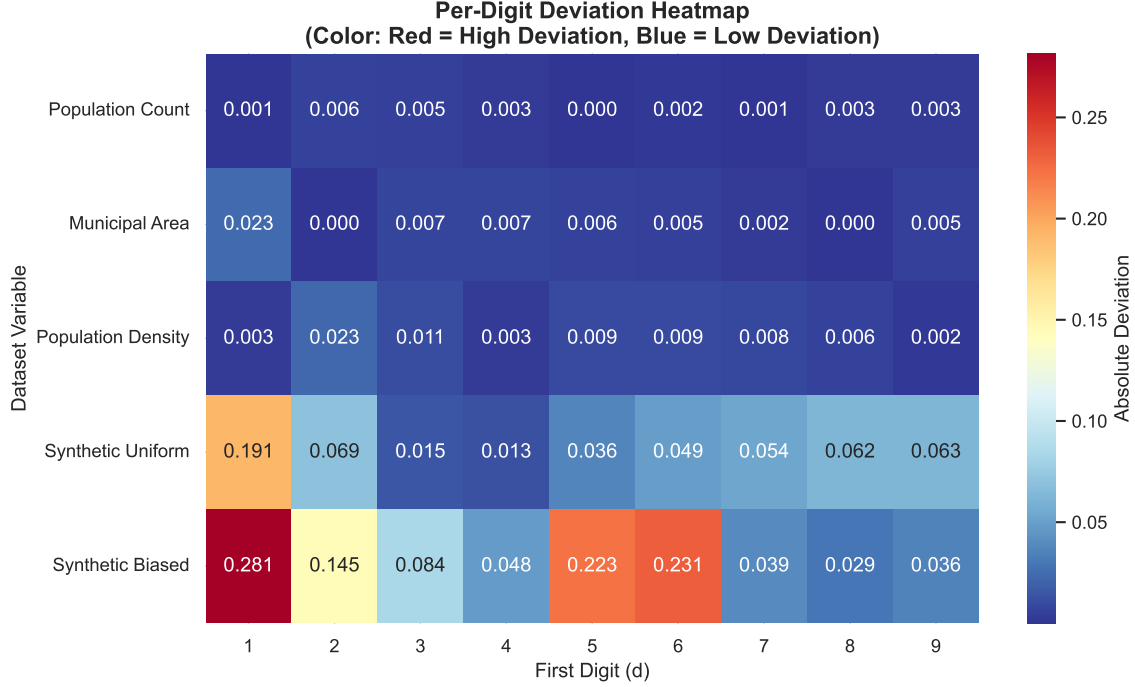


Figure 2: Per-digit absolute deviation heatmap comparing German Zensus 2022 municipal data against synthetic controls. Population Count and Municipal Area exhibit low deviations (<0.03) across all digits, whereas Synthetic Biased displays significant divergence at leading digits ($d=1$: 0.281; $d=5$: 0.223).

integration of synthetic controls allows for a precise calibration of deviation thresholds, confirming that the observed data patterns are consistent with natural generation processes rather than systematic manipulation.

5 Discussion

The analysis of the German Zensus 2022 municipal dataset confirms that population counts, municipal areas, and derived population densities exhibit strong adherence to Benford’s Law. The Mean Absolute Deviation (MAD) scores for these variables remain well below the thresholds defined in Section ??, indicating that leading-digit distributions follow the expected logarithmic pattern [14]. This observation aligns with findings that official demographic statistics often retain statistical signatures of natural processes [1]. The application of Benford’s Law to this high-volume dataset illustrates the method’s capacity to validate large-scale records without relying on subjective claims of success.

A key observation from this study is the divergence between statistical significance and practical relevance in large samples. With approximately 10,800 observations per variable, the Chi-Square test yields strongly significant p-values (e.g., $p < 10^{-7}$) even for variables with excellent conformity [10]. This phenomenon occurs because the Chi-Square statistic scales linearly with sample size, making it overly sensitive to trivial deviations that do not imply data fabrication. Relying solely on p-values in such contexts can lead to false positives, where natural variation is misinterpreted as systematic error or manipulation [16]. By prioritizing the MAD metric, which measures the magnitude of deviation rather than its statistical significance, this study provides a more robust framework for interpreting conformity. The MAD values clearly distinguish between the low-deviation real-world data and the high-deviation synthetic controls, validating the use of proportion-based metrics as a primary diagnostic tool in large-sample

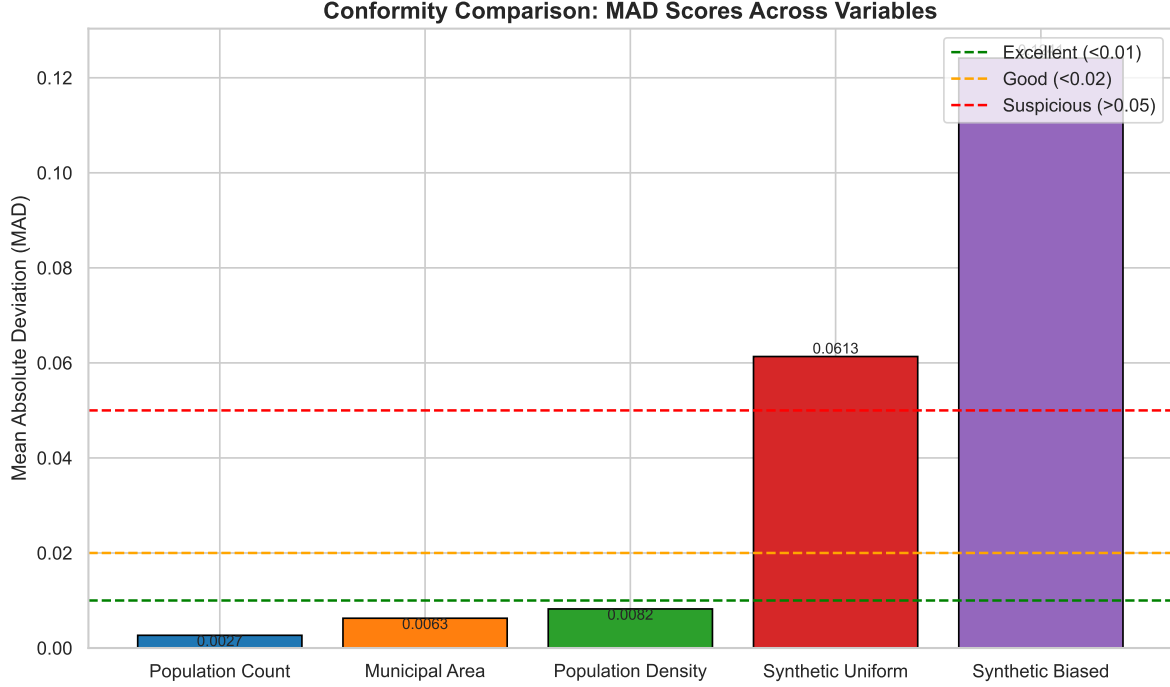


Figure 3: Comparison of Mean Absolute Deviation (MAD) scores for Benford’s Law conformity across German Zensus 2022 variables and synthetic controls. Population Count (0.0027), Municipal Area (0.0063), and Population Density (0.0082) exhibit low deviation values consistent with theoretical expectations, whereas Synthetic Uniform (0.0613) and Synthetic Biased (0.1241) datasets demonstrate significantly higher deviations indicative of anomalous distributions.

forensic analytics [2].

The results also highlight the resilience of Benford’s Law properties in derived variables. Population density is calculated as the ratio of population count to municipal area. Theoretically, operations such as division can distort digit distributions and break the logarithmic pattern expected under Benford’s Law [5]. However, the observed MAD for population density remains low, suggesting that the underlying distributions of the numerator and denominator are sufficiently robust to preserve the overall conformity. This finding implies that derived metrics in official statistics can serve as reliable indicators of data integrity, provided the source variables themselves adhere to natural scaling laws. The synthetic controls further reinforce this conclusion; the uniform and biased datasets exhibit MAD values an order of magnitude higher than the real data, confirming that the observed patterns are not artifacts of the testing procedure but reflect genuine adherence to Benford’s Law [7].

Despite these positive findings, several limitations must be acknowledged. First, the analysis is restricted to a single country (Germany) and a single census year (2022). While this provides a robust validation for the German context, it limits the generalizability of the results to other administrative systems or time periods [3]. Different data collection methodologies, cultural factors in number reporting, or temporal shifts in demographic trends could alter digit distributions. For instance, studies on emission reduction claims have shown that conformity varies by country and dataset characteristics, with some national contexts exhibiting substantial information loss when approximating observed distributions [10]. Second, the synthetic controls used for calibration are simplistic. The uniform distribution assumes complete randomness, and the biased control models only a specific type of human anchoring bias (overrepresentation of 5 and 6). Real-world data fabrication may involve more complex patterns that these baselines do not capture [6]. Consequently, while the current framework effectively distinguishes between natural

data and obvious anomalies, it may lack sensitivity to sophisticated manipulation techniques. Furthermore, Benford’s Law has known limitations in detecting certain manipulations like uniform scaling or in datasets with defined minimums and maximums, which can lead to type-I and type-II errors [10].

Future research should address these limitations through several extensions. Longitudinal analysis comparing Zensus 2022 with previous census years (e.g., Zensus 2011) would allow for the detection of temporal shifts in conformity that might indicate changes in data collection practices or emerging anomalies [14]. Cross-country comparisons could further test the universality of Benford’s Law adherence across different administrative cultures and statistical systems, acknowledging that applicability may vary based on specific national contexts. Additionally, applying this methodology to financial datasets would expand the forensic utility of the approach beyond demographics, potentially offering new tools for detecting irregularities in economic reporting [11]. Finally, refining the synthetic control models to include more complex manipulation scenarios or using machine learning techniques to identify subtle anomalies could enhance the sensitivity of the detection framework [12].

In conclusion, this study demonstrates that Benford’s Law serves as a valid and effective forensic tool for validating the integrity of large-scale official census data. The integration of MAD metrics with synthetic controls provides a nuanced approach to interpreting conformity in high-volume datasets, mitigating the pitfalls of excessive statistical power inherent in traditional Chi-Square tests. The strong adherence of population counts, areas, and densities to Benford’s Law suggests that these variables are generated through natural processes rather than systematic manipulation. While limitations regarding scope and control complexity exist, the proposed methodology offers a solid foundation for future forensic analytics in demographic and administrative statistics.

6 Conclusion

This study confirms that Benford’s Law serves as a robust forensic tool for validating large-scale official census data, such as the German Zensus 2022 municipal records. The analysis demonstrates that population counts and municipal areas exhibit excellent conformity to the expected logarithmic distribution with low Mean Absolute Deviation (MAD) values, while the derived variable Population Density retains this natural pattern even after mathematical operations [5]. Using MAD instead of Chi-Square tests avoids flagging valid data as anomalous due to large sample sizes, providing a reliable indicator for distinguishing between natural variation and systematic manipulation in high-volume datasets [10]. For statistical agencies and auditors, these results suggest that first-digit frequency analysis offers a scalable, automated method for routine quality assurance without relying on subjective judgment.

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